Question	Answer	Marks	Partial Marks
1(a)(i)	Image at (–5, 3), (–1, 3), (–1, 5)	**2**	**B1** for translation $\begin{pmatrix} -7 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 1 \end{pmatrix}$
1(a)(ii)	Translation $\begin{pmatrix} 7 \\ -1 \end{pmatrix}$ cao	**1**	
1(b)	Image at (6, 4), (6, 6), (2, 6)	**2**	**B1** for reflection in line $x = 4$ or for reflection in line $y = k$
1(c)	Image at (2, –2), (2, –6), (4, –6)	**2**	**B1** for correct size and orientation or for rotation 90˚ anticlockwise about (0, 0)
1(d)(i)	Image at (–1, –1), (–3, –1), (–3, –2)	**2**	**B1** for correct size and orientation or for enlargement SF $\frac{1}{2}$, centre (0, 0)
1(d)(ii)	Enlargement and [centre] (0, 0) [factor] –2	**2**	**B1** for Enlargement **and** [centre] (0, 0) **B1** for [factor] –2
2(a)(i)	5	**1**	
2(a)(ii)	17	**1**	
2(a)(iii)	18	**1**	
2(a)(iv)	17.88	**3**	**M2** for (1×15 + 3×16 + 19×17 + 11×18 + 10×19 + 6×20) ÷ 50 oe or **M1** for 1×15 + 3×16 + 19×17 + 11×18 + 10×19 + 6×20 oe
2(b)(i)	1 | 8 2 | 0 0 1 1 1 2 5 3 | 0 3 4	**2**	**B1** for two rows correct or for fully correct unordered stem-and-leaf diagram
2(b)(ii)	21	**1**	
2(b)(iii)	10 nfww	**2**	**B1** for [upper qtile] = 30 or [lower qtile] = 20 soi
3(a)(i)	6925.5[0] cao	**2**	**M1** for $7695 \times \frac{100-10}{100}$ oe or **B1** for answer 769.5
3(a)(ii)	8550	**2**	**M1** for $X \times \frac{100-10}{100} = 7695$ oe

Question	Answer	Marks	Partial Marks
3(b)	660	**3**	**B2** for 60 or **M2** for 600 + $\frac{600\times2\times5}{100}$ oe or **M1** for $\frac{600\times2[\times5]}{100}$ oe
3(c)	1.55 or 1.549 to 1.550	**3**	**M2** for $\sqrt[12]{\frac{601.35}{500}}$ or **M1** for $500\times(...)^{12}=601.35$
3(d)(i)	26.3 or 26.25 to 26.26	**2**	**M1** for $[k]\left(\frac{100-3}{100}\right)^{10}$ oe
3(d)(ii)	23	**3**	**M2** for a correct trial evaluated with n =22 or n = 23 or **M1** for $[k]\ (0.97)^n < 0.5[k]$ oe soi or for $[k](0.97)^n = 0.5[k]$ oe soi, implied by one correct trial $n > 10$ or for $[k](0.97)^{23}$ oe seen If 0 scored **SC1** for answer 22
4(a)	72 or 72.0 cao nfww	**3**	**M2** for $\frac{x}{360}\times2\times\pi\times7.5=2\times\pi\times1.5$ oe or **M1** for $\frac{x}{360}\times2\times\pi\times7.5$ or for $2\times\pi\times1.5$ oe OR **M2** for $\frac{x}{360}\times\pi\times7.5^2=\pi\times1.5\times7.5$ oe or **M1** for $\frac{x}{360}\times\pi\times7.5^2$ or for $\pi\times1.5\times7.5$ oe
4(b)(i)	$2\times\sqrt{17^2-8^2}$ or $\sqrt{34^2-16^2}$ oe	**M2**	**M1** for $17^2=8^2+d^2$ or $34^2 = 16^2 + k^2$
4(b)(ii)	29.3 or 29.30 to 29.31	**4**	**M3** for $([\pi]\times8^2\times30)\div\frac{4}{3}\times[\pi]\times17^3$ [× 100] oe OR **M1** for $\pi\times8^2\times30$ oe **M1** for $\frac{4}{3}\times\pi\times17^3$ oe